

Project APEX

Chapter 4 — Core Research Questions

4.1 Chapter Objective

This chapter defines the primary research questions that guide Project APEX. These questions shape the architecture, experiments, and evaluation criteria for the Human Workflow Learning Platform.

4.2 Primary Research Question

Can an AI system learn a specific user's professional workflow through long-term observation and safely assist or automate that workflow while preserving transparency, privacy, and human oversight?

4.3 Secondary Research Questions

- RQ1.** Which user interactions should be observed?
- RQ2.** How should raw events be transformed into meaningful workflow representations?
- RQ3.** How can reusable skills be extracted from repeated behavior?
- RQ4.** How should user preferences and context be represented?
- RQ5.** How can future user actions be predicted with measurable confidence?
- RQ6.** When should the system assist instead of remaining passive?
- RQ7.** How should autonomy increase while maintaining user control?
- RQ8.** How can continuous feedback improve the behavioral model over time?

4.4 Technical Challenges

- Multi-modal observation (mouse, keyboard, files, applications).
- Context-aware workflow understanding.
- Long-term memory management.
- Explainable decision making.
- Secure local/cloud execution.
- Cross-domain generalization.

4.5 Evaluation Metrics

Success will be evaluated using workflow prediction accuracy, skill extraction quality, user acceptance rate, reduction in repetitive effort, execution reliability, explainability, and privacy compliance.

4.6 Expected Research Contributions

1. Observation-driven AI architecture.
2. Behavioral workflow representation.
3. Human Workflow Learning methodology.
4. Progressive autonomy framework.
5. General-purpose plugin architecture.

4.7 Chapter Summary

The questions defined in this chapter provide the research roadmap for subsequent chapters covering observation, memory, behavior modeling, planning, and execution.